

ANTHONY GRAHAM JR.

MECHANICAL ENGINEERING INTERNSHIP CANDIDATE

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Portfolio: anthony2224.github.io/portfolio

Mechanical Engineering student at LSU with hands-on experience in mechanical design, CAD modeling, HVAC systems, and automation environments. Seeking a mechanical engineering internship.

01 · TECHNICAL SKILLS

CAD & DESIGN	SolidWorks, AutoCAD, Fusion 360 — assemblies, engineering drawings, GD&T
MECHANICAL ENGINEERING	Mechanical disassembly, reverse engineering, form/fit/function analysis, mechanical systems
AUTOMATION & CONTROLS	Conveyor systems, automated material handling, autonomous sorting, control wiring
THERMAL & HVAC SYSTEMS	HVAC controls, facilities HVAC systems, heat pumps, VFDs, system troubleshooting
SOFTWARE & ANALYSIS	MATLAB, Microsoft Excel, Microsoft Word
ENGINEERING METHODS	Design validation, technical reports, root-cause analysis, system optimization

02 · ENGINEERING PROJECTS

Reverse Engineering — Electric Tire Pump ME 1212 · LSU · Team of 6

- Disassembled and analyzed an electromechanical tire pump — components, airflow paths, and system performance.
- Created 3D CAD models and AutoCAD drawings of internal components; proposed design improvements in a structured technical report.

Mechanical Clamp — CAD Design Package CM 1020 · LSU · Team of 2

- Produced a complete SolidWorks package: dimensioned drawings, exploded views, bill of materials, and assembly documentation.
- Analyzed manufacturing methods and material costs; applied GD&T to ANSI Y14.5 drafting standards.

Manufacturing Processes — Casting & Fabrication in Aircraft Engines ME 3633 · LSU · Team of 6

- Researched machining technologies used in aircraft-engine production; mapped component families to casting vs. fabrication methods.
- Documented advantages, disadvantages, and selection trade-offs, illustrated with models and figures.

03 · PROFESSIONAL EXPERIENCE

HVAC Technician — Synergy Energy Management Solutions LLC, Baton Rouge, LA Sep 2024 — Present

- Diagnose, configure, and troubleshoot HVAC control systems — sensors, actuators, electrical components, and automated building systems.
- Support heat pumps, compressors, and variable frequency drives (VFDs); test and adjust systems to improve performance.
- Experience relevant to automotive thermal systems, robotics cooling, and industrial automation.

Fulfillment Associate — Automation Environment — Amazon, Port Allen, LA May — Dec 2024

- Worked a high-volume production environment to standardized processes, quality requirements, and efficiency metrics.
- Assisted system and mechanical assembly operations and building-automation issue resolution.
- Gained exposure to large-scale automation, robotics integration, and process optimization.

04 · EDUCATION

Louisiana State University — Candidate for B.S. in Mechanical Engineering Expected May 2028

Baton Rouge, LA · Member, National Society of Black Engineers (NSBE) — LSU chapter